a) $x[m] = \cos(w_0 . m . T_s)$

Propiedad $\rightarrow$ $x[m] * \delta[m] = x[m]$

$x[m] * \delta[m-n] = x[m-n]$.

$\Rightarrow y[m] = \cos(w_0 . m . T_s) * (\delta[m] - \delta[m-4])$

$y[m] = \cos(w_0 . m . T_s) - \cos(w_0 (m-4) T_s)$

$y[m] = \cos(w_0 . m . T_s) - \cos(w_0 m . T_s - 4 w_0 . T_s)$

Propiedad trigonométrica $\rightarrow$ $\cos\alpha - \cos\beta = -2 \operatorname{sen}\left(\frac{\alpha+\beta}{2}\right) . \operatorname{sen}\left(\frac{\alpha-\beta}{2}\right)$

En este caso, $\alpha = w_0 m . T_s$

$\beta = w_0 . m . T_s - 4 w_0 . T_s$

$\frac{\alpha+\beta}{2} = \frac{2 w_0 . m . T_s - 4 w_0 . T_s}{2} = w_0 . m . T_s - 2 w_0 . T_s$

$\frac{\alpha-\beta}{2} = \frac{4 w_0 T_s}{2} = 2 w_0 . T_s$.

$\Rightarrow y[m] = -2 \operatorname{sen}(w_0 . m T_s - 2 w_0 . T_s) . \operatorname{sen}(2 w_0 . T_s)$

Para que me quede en forma de coseno; uso: $-\operatorname{sen}(\theta) = \cos\left(\theta + \frac{\pi}{2}\right)$

$$y[m] = 2 \operatorname{sen}(2 w_0 . T_s) . \cos\left(w_0 . m . T_s - 2 w_0 . T_s + \frac{\pi}{2}\right)$$

$A = 2 \operatorname{sen}(2 w_0 . T_s)$

$\varphi = -2 w_0 . T_s + \frac{\pi}{2}$

$$y[m] = A \cos(w_0 . m . T_s + \varphi)$$

Escaneado con CamScanner

b) $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$y[n] = x[n] - x[n-4]$.

$y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$.

c) $x[n] = u[n+1] - u[n-2]$.

$y[n] = x[n] - x[n-4]$.

$y[n] = u[n+1] - u[n-2] - (u[n+1-4] - u[n-4-2])$

$y[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6]$.

Escaneado con CamScanner